MeowMemo Plan

AI-assisted cat vocalization interpreter. Audience: students and young professionals. Scope: one-week MVP. October 2026.

1. Product thesis

Build a personalized meow journal, not a literal cat-to-English translator. Make recording effortless and delightful; identify sound types and suggest possible situations; ask the owner to confirm context; learn from saved examples. A joke caption is entertainment, never a verified translation.

2. Target user and core flow

Target: students and young professionals who love their cats, share pet clips with friends, and want a quick "What was that sound?" explanation without complex setup.

Core flow: choose cat, record or upload 2-15 s, detect cat sound, show sound category and tentative context, confirm or edit, save to journal, optionally share a fun card.

3. Competitive landscape

Product	Features observed	Opportunity for MeowMemo
MeowTalk (original), meowtalk.app	AI-based 11 broad intents; recording, correction-based per-cat training, and premium intent access.	Simpler onboarding; transparent uncertainty; clear labels and corrections; no paywall for the core journal.
MeowTalk-named / cat-translator entertainment apps (App Store listings)	Human-to-cat playful sounds, soundboard, games, whistle sounds, replay, and subscriptions.	Keep fun share cards, but clearly separate simulated dialogue and sound effects from AI inference.
MeowTalk.io (promotional product site)	Claims real-time translation, 12 emotion labels, games, whistles, and multilingual use.	Avoid unvalidated emotion or accuracy promises; offer sound type + context + "unsure."

Competitor product claims are their own claims, not independently validated performance.

4. Open research datasets

Dataset	Value for the prototype	Cautions / permission
CatMeows (zenodo.org/records/4008297)	440 meows from 21 cats; labels for food waiting, brushing, and unfamiliar-room isolation; filenames include cat ID.	Research/noncommercial only; contextual conditions are not emotion ground truth. Use cat-held-out splits.
ESC-50 (github.com/karolpiczak/ESC-50)	2,000 5-s environmental sounds; cat class and negative classes useful for initial cat-vs-other screening.	CC BY-NC for the full dataset; cat class has only 40 clips, insufficient for nuanced interpretation.
FSD50K (zenodo.org/records/4060432)	51,197 sound-event clips with 200 labels; additional cat and environmental examples.	Mixed per-clip CC licenses; filter and track attribution before reuse. Not an intent-labeled meow set.

5. MVP requirements and acceptance tests

Scope: one mobile-first web app with real audio handling and a narrow, honest inference pathway.

Priority	Requirement / definition of done
P0 - Record	Record or upload 2-15 s audio; permission and error states; playback and visible waveform; works on mobile HTTPS.
P0 - Classify	Real audio model or API screens "cat vocalization likely / uncertain / not detected." If using the CatMeows classifier, only suggest its 3 recording contexts and label them as possibilities.
P0 - Explain	Result shows detected sound or "unsure," 1-3 possible contexts, what evidence was used, and a clear non-literal-translation disclaimer. No fabricated numeric confidence.
P0 - Feedback	User picks food / brushing / isolation / other / unknown or adds a note; label is stored with audio, cat, time, and model version.
P0 - Journal	Create cat profile (name + optional photo); revisit, play, and delete audio; filter recordings by cat and user label.
P1 - Share	One-tap image card with cat name, waveform, and playful interpretation clearly labeled "for fun"; nothing public by default.
P2 - Later	Live always-on listening, personalized retraining, more emotion categories, speech synthesis, health diagnosis, and paid plans.

6. Work items

All items fit within one week of work. Order and pacing are flexible; each item has a testable output.

Work item	Testable output
Scaffold Next.js + TypeScript + Tailwind; define cat and recording types.	Responsive 3-screen prototype with functional navigation.
Browser recorder, upload, trim/playback, and waveform.	Record or upload on a phone; silence and error states handled.
Dataset audit and baseline: CatMeows 3-context split by cat; ESC-50 negative sound screening.	Documented baseline metrics on unseen cats; working "unknown" path.
Python FastAPI inference endpoint and results UI.	Uploaded clip returns actual model output, never a mock result disguised as AI.
Cat profiles, local or Supabase storage, correction form, and journal.	Corrections persist; audio can be replayed and deleted.
Optional share card, polish, accessibility, and privacy consent.	Sharing requires user action; clear demo and uncertainty labels.
Pilot with 3-5 cat owners; fix recording, noise, and prediction errors.	Deployed HTTPS demo, README, limitations, and a 2-minute demo video.

7. Architecture and guardrails

- Stack: Next.js / TypeScript / Tailwind; MediaRecorder Web API; Python + librosa/scikit-learn or pretrained audio embeddings; FastAPI inference; Supabase or browser-local storage for a fast pilot.
- A language model may word a result, but must not fabricate a diagnosis or determine classes without real acoustic input.
- Validation: split CatMeows by cat ID, not random clips; report context macro-F1 and a confusion matrix; test silence, dog/human sounds, and noise.
- A 3-class dataset is not evidence of understanding emotions. Offer "unknown" whenever evidence is insufficient.
- Privacy and welfare: microphone consent, deletion, opt-in data sharing; never infer pain or illness, and never play distressing sounds to provoke recordings.

8. Success criteria

- Ship: public demo + code + model card.
- Usability: 4 of 5 pilot users complete a first recording without help.
- Integrity: every model result is traceable to audio inference; uncertain audio abstains; no unsupported accuracy claims.
- Next research step: test whether individual-cat labeled examples improve prediction over the generic baseline.

Sources

meowtalk.app; meowtalk.app/faqs; apps.apple.com (OYE TEKNOLOJI listing); meowtalk.io/en; CatMeows (Zenodo); ESC-50 (GitHub); FSD50K (Zenodo).